

Cluster Analysis: Applications

- ❑ A key intermediate step for other data mining tasks
 - ❑ Generating a compact summary of data for classification, pattern discovery, hypothesis generation and testing, etc.
 - ❑ Outlier detection: Outliers—those “far away” from any cluster
- ❑ Data summarization, compression, and reduction
 - ❑ Ex. Image processing: Vector quantization
- ❑ Collaborative filtering, recommendation systems, or customer segmentation
 - ❑ Find like-minded users or similar products
- ❑ Dynamic trend detection
 - ❑ Clustering stream data and detecting trends and patterns
- ❑ Multimedia data analysis, biological data analysis and social network analysis
 - ❑ Ex. Clustering images or video/audio clips, gene/protein sequences, etc.

Considerations for Cluster Analysis

❑ Partitioning criteria

- ❑ Single level vs. hierarchical partitioning (often, multi-level hierarchical partitioning is desirable, e.g., grouping topical terms)

❑ Separation of clusters

- ❑ Exclusive (e.g., one customer belongs to only one region) vs. non-exclusive (e.g., one document may belong to more than one class)

❑ Similarity measure

- ❑ Distance-based (e.g., Euclidean, road network, vector) vs. connectivity-based (e.g., density or contiguity)

❑ Clustering space

- ❑ Full space (often when low dimensional) vs. subspaces (often in high-dimensional clustering)